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### Exercise 1.3

**Q1.** Factorize the following:

(i)  $a^2 + 4b^2$

Sol.  $a^2 + 4b^2$

$$= a^2 + 4b^2 (-1)$$

$$= a^2 - 4b^2 i^2$$

$$\because i^2 = -1$$

$$= (a)^2 - (2bi)^2$$

$$= (a + 2bi)(a - 2bi)$$

(ii)  $9a^2 + 16b^2$

Sol.  $9a^2 + 16b^2$   
 $= 9a^2 - 16b^2(-1)$   
 $= 9a^2 - 16b^2i^2$   
 $= (3a)^2 - (4bi)^2$   
 $= (3a + 4bi)(3a - 4bi)$

$\because i^2 = -1$

(iii)  $3x^2 + 3y^2$

Sol.  $3x^2 + 3y^2$   
 $= (x^2 + y^2)$   
 $= 3[x^2 - y^2(-1)]$   
 $= 3[x^2 - y^2i^2]$   
 $= 3[(x)^2 - (yi)^2]$   
 $= 3(x + yi)(x - yi)$

$\because i^2 = -1$

(iv)  $144x^2 + 225y^2$

Sol.  $144x^2 + 225y^2$   
 $= 144x^2 - 225y^2(-1)$   
 $= 144x^2 - 225y^2i^2$   
 $= (12x)^2 - (15yi)^2$   
 $= (12x + 15yi)(12x - 15yi)$

$\because i^2 = -1$

(v)  $z^2 - 2iz - 1$

Sol.  $z^2 - 2iz - 1$   
 $= z^2 - 2iz + (-1)$   
 $= z^2 - 2iz + i^2$   
 $= z^2 - 2(z)(i) + i^2$   
 $= (z - i)^2$   
 $= (z - i)(z - i)$

$\because i^2 = -1$

(vi)  $z^2 + 6z + 13$

Sol.  $z^2 + 6z + 13$   
 $= z^2 + 6z + 9 + 4$   
 $= z^2 + 2(z)(3) + 3^2 - 4(-1)$   
 $= (z + 3)^2 - 4i^2$   
 $= (z + 3)^2 - (2i)^2$   
 $= (z + 3 + 2i)(z + 3 - 2i)$

$\because i^2 = -1$

(vii)  $z^2 + 4z + 5$

Sol.  $z^2 + 4z + 5$   
 $= z^2 + 4z + 4 + 1$   
 $= z^2 + 2(z)(2) + 2^2 - (-1)$   
 $= (z + 2)^2 - i^2$   
 $= (z + 2 + i)(z + 2 - i)$

$\because i^2 = -1$

(viii)  $2z^2 - 22z + 65$

Sol.  $2z^2 - 22z + 65$   
 $= 2\left(z^2 - 11z + \frac{65}{2}\right)$   
 $= 2\left[z^2 - 2(z)\left(\frac{11}{2}\right) + \left(\frac{11}{2}\right)^2 - \left(\frac{11}{2}\right)^2 + \frac{65}{2}\right]$

$$= 2\left[\left(z - \frac{11}{2}\right)^2 - \frac{121}{4} + \frac{65}{2}\right]$$

$$= 2\left[\left(z - \frac{11}{2}\right)^2 - \left(\frac{121}{4} - \frac{65}{2}\right)\right]$$

$$= 2\left[\left(z - \frac{11}{2}\right)^2 - \left(\frac{121 - 130}{4}\right)\right]$$

$$= 2\left[\left(z - \frac{11}{2}\right)^2 - \left(\frac{-9}{4}\right)\right]$$

$$= 2\left[\left(z - \frac{11}{2}\right)^2 - \left(\frac{9i^2}{4}\right)\right]$$

$\because i^2 = -1$

$$= 2\left[\left(z - \frac{11}{2}\right)^2 - \left(\frac{3i}{2}\right)^2\right]$$

$$= 2\left(z - \frac{11}{2} + \frac{3i}{2}\right)\left(z - \frac{11}{2} - \frac{3i}{2}\right)$$

$$= 2\left[z - \left(\frac{11}{2} - \frac{3i}{2}\right)\right]\left[z - \left(\frac{11}{2} + \frac{3i}{2}\right)\right]$$

$$= 2\left(z - \frac{11 - 3i}{2}\right)\left(z - \frac{11 + 3i}{2}\right)$$

**02** Factorize the following polynomials into its linear factors:

(i)  $z^3 + 8$

Sol.  $z^3 + 8$   
 $= z^3 + 2^3$   
 $= (z + 2)(z^2 - 2z + 4)$   
 $= (z + 2)(z^2 - 2z + 1 + 3)$   
 $= (z + 2)[z^2 - 2(z)(1) + 1^2 - 3(-1)]$   
 $= (z + 2)[(z - 1)^2 - 3i^2]$   
 $= (z + 2)\left[(z - 1)^2 - (\sqrt{3}i)^2\right]$   
 $= (z + 2)(z - 1 + \sqrt{3}i)(z - 1 - \sqrt{3}i)$

(ii)  $z^3 + 27$

Sol.  $z^3 + 27$   
 $= z^3 + 3^3$   
 $= (z + 3)(z^2 - 3z + 9)$   
 $= (z + 3)\left[z^2 - 2(z)\left(\frac{3}{2}\right) + \left(\frac{3}{2}\right)^2 - \left(\frac{3}{2}\right)^2 + 9\right]$   
 $= (z + 3)\left[\left(z - \frac{3}{2}\right)^2 - \frac{9}{4} + 9\right]$



$$\begin{aligned}
 &= (z+3) \left[ \left( z - \frac{3}{2} \right)^2 - \left( \frac{9}{4} - 9 \right) \right] \\
 &= (z+3) \left[ \left( z - \frac{3}{2} \right)^2 - \left( \frac{9-36}{4} \right) \right] \\
 &= (z+3) \left[ \left( z - \frac{3}{2} \right)^2 - \left( \frac{-27}{4} \right) \right] \\
 &= (z+3) \left[ \left( z - \frac{3}{2} \right)^2 - \left( \frac{27i^2}{4} \right) \right] \quad \because i^2 = -1 \\
 &= (z+3) \left[ \left( z - \frac{3}{2} \right)^2 - \left( \frac{3\sqrt{3}i}{2} \right)^2 \right] \\
 &= (z+3) \left( z - \frac{3}{2} + \frac{3\sqrt{3}i}{2} \right) \left( z - \frac{3}{2} - \frac{3\sqrt{3}i}{2} \right) \\
 &= (z+3) \left[ z - \left( \frac{3}{2} - \frac{3\sqrt{3}i}{2} \right) \right] \left[ z - \left( \frac{3}{2} + \frac{3\sqrt{3}i}{2} \right) \right] \\
 &= (z+3) \left( z - \frac{3-3\sqrt{3}i}{2} \right) \left( z - \frac{3+3\sqrt{3}i}{2} \right)
 \end{aligned}$$

(iii)  $z^3 - 2z^2 + 16z - 32$

Sol.  $z^3 - 2z^2 + 16z - 32$

$$\begin{aligned}
 &= z^2(z-2) + 16(z-2) \\
 &= (z-2)(z^2 + 16) \\
 &= (z-2)[z^2 - 16(-1)] \\
 &= (z-2)(z^2 - 16i^2) \quad \because i^2 = -1 \\
 &= (z-2)[z^2 - (4i)^2] \\
 &= (z-2)(z+4i)(z-4i)
 \end{aligned}$$

(iv)  $z^4 + 21z^2 - 100$

Sol.  $z^4 + 21z^2 - 100$

$$\begin{aligned}
 &= z^4 + 25z^2 - 4z^2 - 100 \\
 &= z^2(z^2 + 25) - 4(z^2 + 25) \\
 &= (z^2 + 25)(z^2 - 4) \\
 &= [z^2 - 25(-1)](z^2 - 2^2) \\
 &= (z^2 - 25i^2)(z+2)(z-2) \quad \because i^2 = -1 \\
 &= [z^2 - (5i)^2](z+2)(z-2) \\
 &= (z+5i)(z-5i)(z+2)(z-2)
 \end{aligned}$$

(v)  $z^4 - 16$

Sol.  $z^4 - 16$

$$\begin{aligned}
 &= (z^2)^2 - 4^2 \\
 &= (z^2 + 4)(z^2 - 4) \\
 &= [z^2 - 4(-1)](z^2 - 2^2) \\
 &= (z^2 - 4i^2)(z+2)(z-2) \quad \because i^2 = -1 \\
 &= [z^2 - (2i)^2](z+2)(z-2) \\
 &= (z+2i)(z-2i)(z+2)(z-2)
 \end{aligned}$$

(vi)  $z^4 + 3z^2 - 4$

Sol.  $z^4 + 3z^2 - 4$

$$\begin{aligned}
 &= z^4 + 4z^2 - z^2 - 4 \\
 &= z^2(z^2 + 4) - 1(z^2 + 4) \\
 &= (z^2 + 4)(z^2 - 1) \\
 &= [z^2 - 4(-1)](z^2 - 1^2) \\
 &= (z^2 - 4i^2)(z+1)(z-1) \quad \because i^2 = -1 \\
 &= [z^2 - (2i)^2](z+1)(z-1) \\
 &= (z+2i)(z-2i)(z+1)(z-1)
 \end{aligned}$$

(vii)  $z^4 + 5z^2 + 6$

Sol.  $z^4 + 5z^2 + 6$

$$\begin{aligned}
 &= z^4 + 3z^2 + 2z^2 + 6 \\
 &= z^2(z^2 + 3) + 2(z^2 + 3) \\
 &= (z^2 + 3)(z^2 + 2) \\
 &= [z^2 - 3(-1)][z^2 - 2(-1)] \\
 &= (z^2 - 3i^2)(z^2 - 2i^2) \quad \because i^2 = -1 \\
 &= [z^2 - (\sqrt{3}i)^2][z^2 - (\sqrt{2}i)^2] \\
 &= (z + \sqrt{3}i)(z - \sqrt{3}i)(z + \sqrt{2}i)(z - \sqrt{2}i)
 \end{aligned}$$

(viii)  $z^4 - 32z^2 - 3969$

Sol.  $z^4 - 32z^2 - 3969$

$$\begin{aligned}
 &= z^4 - 81z^2 + 49z^2 - 3969 \\
 &= z^2(z^2 - 81) + 49(z^2 - 81) \\
 &= (z^2 - 81)(z^2 + 49) \\
 &= (z^2 - 81)(z^2 - 49(-1)) \\
 &= (z^2 - 81)(z^2 - 49i^2) \\
 &= (z^2 - (9)^2)(z^2 - (7i)^2) \\
 &= (z-9)(z+9)(z-7i)(z+7i)
 \end{aligned}$$

**Q3.** Find the roots of  $z^4 + 7z^2 - 144 = 0$  and hence express it as a product of linear factors.

Sol.  $z^4 + 7z^2 - 144 = 0$

$$\begin{aligned}
 &z^4 + 16z^2 - 9z^2 - 144 = 0 \\
 &z^2(z^2 + 16) - 9(z^2 + 16) = 0 \\
 &(z^2 + 16)(z^2 - 9) = 0
 \end{aligned}$$

Either

$$\begin{aligned}
 z^2 + 16 &= 0 \\
 z^2 - 16i^2 &= 0 \\
 z^2 - (4i)^2 &= 0 \\
 (z+4i)(z-4i) &= 0 \\
 z+4i &= 0 \text{ or } z-4i = 0 \\
 z &= -4i \text{ or } z = 4i
 \end{aligned}$$

Or

$$\begin{aligned}
 z^2 - 9 &= 0 \\
 z^2 - 3^2 &= 0 \\
 (z+3)(z-3) &= 0 \\
 z+3 &= 0 \text{ or } z-3 = 0 \\
 z &= -3 \text{ or } z = 3
 \end{aligned}$$

By using the corollary of fundamental theorem of Algebra, the equation can be factorized using the roots

$$z^4 + 7z^2 - 144 = (z+4i)(z-4i)(z+3)(z-3)$$



**Q4.** Solve the following complex quadratic equations by completing square method:

(i)  $2z^2 - 3z + 4 = 0$

Sol.  $2z^2 - 3z + 4 = 0$

Divide by 2 on both side

$$z^2 - \frac{3}{2}z + 2 = 0$$

$$z^2 - \frac{3}{2}z = -2$$

$$z^2 - \frac{3}{2}z + \left(\frac{3}{4}\right)^2 = -2 + \left(\frac{3}{4}\right)^2$$

$$z^2 - 2(z)\left(\frac{3}{4}\right) + \left(\frac{3}{4}\right)^2 = -2 + \frac{9}{16}$$

$$\left(z - \frac{3}{4}\right)^2 = \frac{-32 + 9}{16}$$

$$\left(z - \frac{3}{4}\right)^2 = \frac{-23}{16}$$

$$\sqrt{\left(z - \frac{3}{4}\right)^2} = \pm \sqrt{\frac{-23}{16}}$$

$$z - \frac{3}{4} = \pm \frac{\sqrt{23}i}{4}$$

$$z = \frac{3}{4} \pm \frac{\sqrt{23}i}{4}$$

$$z = \frac{3 \pm \sqrt{23}i}{4}$$

(ii)  $z^2 - 6z + 30 = 0$

Sol.  $z^2 - 6z = -30$

Add  $3^2$  on both side

$$z^2 - 6z + 3^2 = -30 + 3^2$$

$$z^2 - 2(z)(3) + 3^2 = -30 + 9$$

$$(z - 3)^2 = -21$$

$$\sqrt{(z - 3)^2} = \pm \sqrt{-21}$$

$$z - 3 = \pm \sqrt{21}i$$

$$z = 3 \pm \sqrt{21}i$$

(iii)  $3z^2 - 18z + 50 = 0$

Sol.  $3z^2 - 18z + 50 = 0$

Divide by 3 on both sides

$$z^2 - 6z + \frac{50}{3} = 0$$

$$z^2 - 6z = -\frac{50}{3}$$

$$z^2 - 6z + 3^2 = -\frac{50}{3} + 3^2$$

$$z^2 - 2(z)(3) + 3^2 = -\frac{50}{3} + 9$$

$$(z - 3)^2 = \frac{-50 + 27}{3}$$

$$(z - 3)^2 = \frac{-23}{3}$$

$$\sqrt{(z - 3)^2} = \pm \sqrt{\frac{-23}{3}}$$

$$z - 3 = \pm \sqrt{\frac{23}{3}}i$$

$$z = 3 \pm \sqrt{\frac{23}{3}}i$$

$$z = 3 \pm \frac{\sqrt{23}}{\sqrt{3}}i$$

$$z = \frac{3\sqrt{3} \pm \sqrt{23}i}{\sqrt{3}}$$

(iv)  $z^2 + 4z + 13 = 0$

Sol.  $z^2 + 4z = -13$

Add  $2^2$  on both side

$$z^2 + 4z + 2^2 = -13 + 2^2$$

$$z^2 + 2(z)(2) + 2^2 = -13 + 4$$

$$(z + 2)^2 = -9$$

$$\sqrt{(z + 2)^2} = \pm \sqrt{-9}$$

$$z + 2 = \pm 3i$$

$$z = -2 \pm 3i$$

(v)  $2z^2 + 6z + 9 = 0$

Sol.  $2z^2 + 6z + 9 = 0$

Divide by 2 on both side

$$z^2 + 3z + \frac{9}{2} = 0$$

$$z^2 + 3z = -\frac{9}{2}$$

$$z^2 + 3z + \left(\frac{3}{2}\right)^2 = -\frac{9}{2} + \left(\frac{3}{2}\right)^2$$

$$z^2 + 2(z)\left(\frac{3}{2}\right) + \left(\frac{3}{2}\right)^2 = -\frac{9}{2} + \frac{9}{4}$$

$$\left(z + \frac{3}{2}\right)^2 = \frac{-18 + 9}{4}$$

$$\left(z + \frac{3}{2}\right)^2 = \frac{-9}{4}$$

$$\sqrt{\left(z + \frac{3}{2}\right)^2} = \pm \sqrt{\frac{-9}{4}}$$



$$z + \frac{3}{2} = \pm \frac{3i}{2}$$

$$z = -\frac{3}{2} \pm \frac{3i}{2}$$

$$z = \frac{-3 \pm 3i}{2}$$

(vi)  $3z^2 - 5z + 7 = 0$

Sol.  $3z^2 - 5z + 7 = 0$

Divide by 3 on both sides

$$z^2 - \frac{5}{3}z + \frac{7}{3} = 0$$

$$z^2 - \frac{5}{3}z = -\frac{7}{3}$$

Add  $\left(\frac{5}{6}\right)^2$  on both side

$$z^2 - \frac{5}{3}z + \left(\frac{5}{6}\right)^2 = -\frac{7}{3} + \left(\frac{5}{6}\right)^2$$

$$z^2 - 2\left(z\right)\left(\frac{5}{6}\right) + \left(\frac{5}{6}\right)^2 = -\frac{7}{3} + \frac{25}{36}$$

$$\left(z - \frac{5}{6}\right)^2 = \frac{-84 + 25}{36}$$

$$\left(z - \frac{5}{6}\right)^2 = \frac{-59}{36}$$

$$\sqrt{\left(z - \frac{5}{6}\right)^2} = \pm \sqrt{\frac{-59}{36}}$$

$$z - \frac{5}{6} = \pm \frac{\sqrt{59}i}{6}$$

$$z = \frac{5}{6} \pm \frac{\sqrt{59}i}{6}$$

$$z = \frac{5 \pm \sqrt{59}i}{6}$$

**Q5.** Solve the following equations:

(i)  $2z^4 - 32 = 0$

Sol.  $2(z^4 - 16) = 0$

$$z^4 - 16 = 0$$

$$(z^2)^2 - 4^2 = 0$$

$$(z^2 + 4)(z^2 - 4) = 0$$

$$z^2 + 4 = 0$$

$$z^2 - 4i^2 = 0$$

$$z^2 - (2i)^2 = 0$$

$$(z + 2i)(z - 2i) = 0$$

$$z + 2i = 0 \text{ or } z - 2i = 0$$

$$z = -2i \text{ or } z = 2i$$

$$z^2 - 4 = 0$$

$$z^2 - 2^2 = 0$$

$$(z + 2)(z - 2) = 0$$

$$z + 2 = 0 \text{ or } z - 2 = 0$$

$$z = -2 \text{ or } z = 2$$

(ii)  $3z^5 - 243z = 0$

Sol.  $3z(z^4 - 81) = 0$

$$\Rightarrow z = 0, z^4 - 81 = 0$$

$$(z^2)^2 - 9^2 = 0$$

$$(z^2 + 9)(z^2 - 9) = 0$$

$$z^2 + 9 = 0$$

$$z^2 - 9i^2 = 0$$

$$z^2 - (3i)^2 = 0$$

$$(z + 3i)(z - 3i) = 0$$

$$z + 3i = 0 \text{ or } z - 3i = 0$$

$$z = -3i \text{ or } z = 3i$$

$$z^2 - 9 = 0$$

$$z^2 - 3^2 = 0$$

$$(z + 3)(z - 3) = 0$$

$$z + 3 = 0 \text{ or } z - 3 = 0$$

$$z = -3 \text{ or } z = 3$$

(iii)  $5z^5 - 5z = 0$

Sol.  $5z(z^4 - 1) = 0$

$$\Rightarrow z = 0, z^4 - 1 = 0$$

$$(z^2)^2 - 1^2 = 0$$

$$(z^2 + 1)(z^2 - 1) = 0$$

$$z^2 + 1 = 0$$

$$z^2 - i^2 = 0$$

$$(z + i)(z - i) = 0$$

$$z + i = 0 \text{ or } z - i = 0$$

$$z = -i \text{ or } z = i$$

$$z^2 - 1 = 0$$

$$z^2 - 1^2 = 0$$

$$(z + 1)(z - 1) = 0$$

$$z + 1 = 0 \text{ or } z - 1 = 0$$

$$z = -1 \text{ or } z = 1$$

(iv)  $z^3 - 5z^2 + z - 5 = 0$

Sol.  $z^2(z - 5) + 1(z - 5) = 0$

$$(z - 5)(z^2 + 1) = 0$$

$$z - 5 = 0$$

$$z = 5$$

$$z^2 + 1 = 0$$

$$z^2 - i^2 = 0$$

$$(z + i)(z - i) = 0$$

$$z + i = 0 \text{ or } z - i = 0$$

$$z = -i \text{ or } z = i$$

(v)  $4z^4 - 25z^2 - 21 = 0$

Sol.  $4z^4 - 25z^2 - 21 = 0$

$$\Rightarrow 4(z^2)^2 - 25z^2 - 21 = 0$$

Here  $a = 1, b = -25, c = -21$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-(-25) \pm \sqrt{(-25)^2 - 4(4)(-21)}}{2(4)}$$

$$= \frac{25 \pm \sqrt{625 + 336}}{8} = \frac{25 \pm \sqrt{961}}{8} = \frac{25 \pm 31}{8}$$

$$x = \frac{25 + 31}{8} = \frac{56}{8}$$

$$x = 7$$

$$\Rightarrow z^2 = 7$$

$$\Rightarrow z = \pm\sqrt{7}$$

$$x = \frac{25 - 31}{8} = \frac{-6}{8}$$

$$x = -\frac{3}{4}$$

$$\Rightarrow z^2 = -\frac{3}{4}$$

$$\Rightarrow z = \pm\sqrt{-\frac{3}{4}}$$

$$z = \pm \frac{i\sqrt{3}}{2}$$



$$(vi) z^3 + z^2 + z + 1 = 0$$

$$\text{Sol. } z^2(z+1) + 1(z+1) = 0$$

$$(z+1)(z^2+1) = 0$$

Either

$$z+1=0$$

$$z=-1$$

Or

$$z^2+1=0$$

$$z^2 - i^2 = 0$$

$$(z+i)(z-i) = 0$$

$$z+i=0 \text{ or } z-i=0$$

$$z=-i \text{ or } z=i$$

**Q6.** Find a polynomial  $P(z)$  of degree 3 with zeros 3,  $-2i$ ,  $2i$  and satisfying  $P(1) = 20$ .

**Sol.** By using the general form of degree 3

$$P(z) = a(z-z_1)(z-z_2)(z-z_3)$$

As, 3,  $-2i$ ,  $2i$  are zeros/roots of the polynomial.

$$\text{So, } P(z) = a(z-3)[z-(-2i)](z-2i)$$

$$P(z) = a(z-3)(z+2i)(z-2i)$$

$$P(z) = a(z-3)[(z)^2 - (2i)^2]$$

$$P(z) = a(z-3)(z^2 - 4i^2)$$

$$P(z) = a(z-3)(z^2 - 4(-1))$$

$$P(z) = a(z-3)(z^2 + 4)$$

$$P(z) = a(z^3 - 3z^2 + 4z - 12) \quad \text{--- (i)}$$

Put  $z = 1$  in (i) to find the value of  $a$

$$P(1) = a[1^3 - 3(1)^2 + 4(1) - 12]$$

$$P(1) = a[1 - 3(1) + 4(1) - 12]$$

$$P(1) = a(1 - 3 + 4 - 12)$$

$$P(1) = a(-10)$$

$$20 = a(-10) \quad \therefore p(1) = 20$$

$$\frac{20}{-10} = a$$

$$a = -2$$

Now, put the value of  $a$  in (i)

$$p(z) = -2(z^3 - 3z^2 + 4z - 12)$$

$$p(z) = -2z^3 + 6z^2 - 8z + 24$$

**Q7.** Find a polynomial  $P(z)$  of degree 4 with zeros  $2i$ ,  $-2i$ ,  $1$ ,  $-1$ , and satisfying  $P(2) = 240$ .

**Sol.** By using the general form of degree 4

$$P(z) = a(z-z_1)(z-z_2)(z-z_3)(z-z_4)$$

As,  $2i$ ,  $-2i$ ,  $1$ ,  $-1$  are zeros/roots of the polynomial. So,

$$P(z) = a(z-2i)[z-(-2i)](z-1)[z-(-1)]$$

$$P(z) = a(z-2i)(z+2i)(z-1)(z+1)$$

$$P(z) = a[(z)^2 - (2i)^2][z^2 - 1^2]$$

$$P(z) = a(z^2 - 4i^2)(z^2 - 1)$$

$$P(z) = a[z^2 - 4(-1)](z^2 - 1)$$

$$P(z) = a(z^2 + 4)(z^2 - 1)$$

$$P(z) = a(z^4 - z^2 + 4z^2 - 4)$$

$$P(z) = a(z^4 + 3z^2 + 4) \quad \text{--- (i)}$$

Put  $z = 2$  in (i) to find the value of  $a$

$$P(2) = a[2^4 + 3(2)^2 + 4]$$

$$P(2) = a[16 + 3(4) + 4]$$

$$P(2) = a(16 + 12 + 4)$$

$$P(2) = a(24)$$

$$240 = a(24) \quad \therefore p(2) = 240$$

$$\frac{240}{24} = a$$

$$a = 10$$

Now, put the value of  $a$  in (i)

$$P(z) = 10(z^4 + 3z^2 + 4)$$

$$P(z) = 10z^4 + 30z^2 + 40$$

**Q8.** Find a polynomial  $P(z)$  of degree 4 with zeros 4,  $-4$ ,  $1+i$ ,  $1-i$  and satisfying  $P(2) = 72$ .

**Sol.** By using the general form of degree 4

$$P(z) = a(z-z_1)(z-z_2)(z-z_3)(z-z_4)$$

As, 4,  $-4$ ,  $1+i$ ,  $1-i$  are zeros/roots of the polynomial. So,

$$P(z) = a(z-4)[z-(-4)][z-(1+i)][z-(1-i)]$$

$$P(z) = a(z-4)(z+4)(z-1-i)(z-1+i)$$

$$P(z) = a(z^2 - 4^2)(z^2 - z + zi - z + 1 - i - zi + i - i^2)$$

$$P(z) = a(z^2 - 16)[z^2 - z - z + 1 - (-1)]$$

$$P(z) = a(z^2 - 16)(z^2 - z - z + 1 + 1)$$

$$P(z) = a(z^2 - 16)(z^2 - 2z + 2)$$

$$P(z) = a(z^4 - 2z^3 + 2z^2 - 16z^2 + 32z - 32)$$

$$P(z) = a(z^4 - 2z^3 - 14z^2 + 32z - 32) \quad \text{--- (i)}$$

Put  $z = 2$  in (i) to find the value of  $a$

$$P(2) = a[2^4 - 2(2)^3 - 14(2)^2 + 32(2) - 32]$$

$$P(2) = a[16 - 2(8) - 14(4) + 32(2) - 32]$$

$$P(2) = a(16 - 16 - 56 + 64 - 32)$$

$$P(2) = a(-24)$$

$$72 = a(-24) \quad \therefore p(2) = 72$$

$$\frac{72}{-24} = a$$

$$a = -3$$

Now, put the value of  $a$  in (i)

$$P(z) = -3(z^4 - 2z^3 - 14z^2 + 32z - 32)$$

$$P(z) = -3z^4 + 6z^3 + 42z^2 - 96z + 96$$

## 1.4 Three Cube Roots of Unity

Let  $x$  be a cube root of unity

$$\therefore x = (1)^{\frac{1}{3}}$$

$$\Rightarrow x^3 = 1$$

$$\Rightarrow x^3 - 1 = 0$$

$$\Rightarrow (x-1)(x^2 + x + 1) = 0$$

$$\text{Either } x-1=0 \Rightarrow x=1$$

$$\text{or } x^2 + x + 1 = 0$$